Grokking the Advanced System Design Interview

Learn system design through architectural review of real systems.

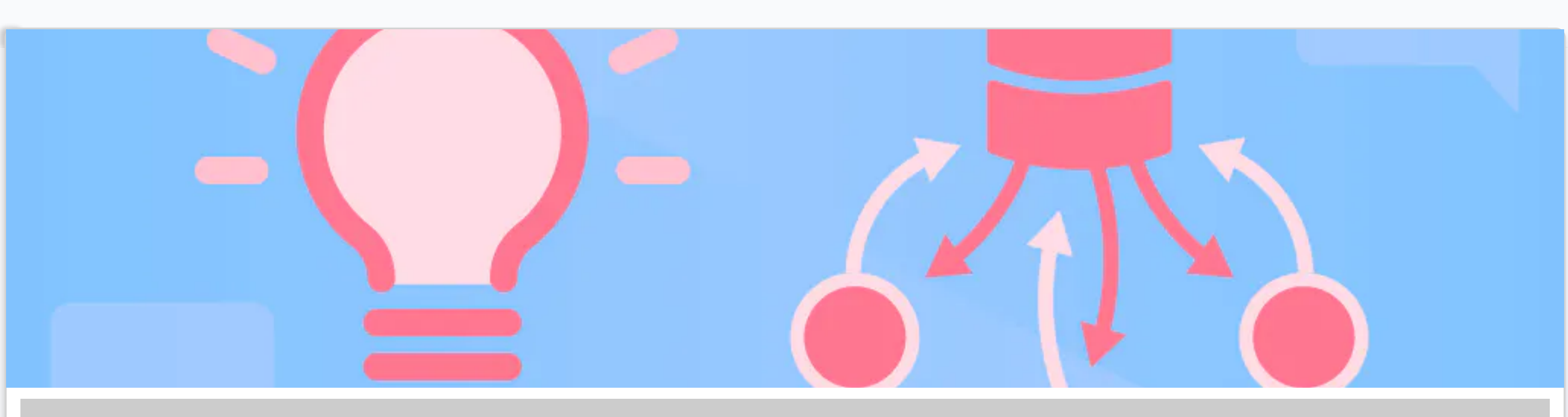
Level: Advanced | Study Time: 21h | Lessons: 118

33,977 (Students)

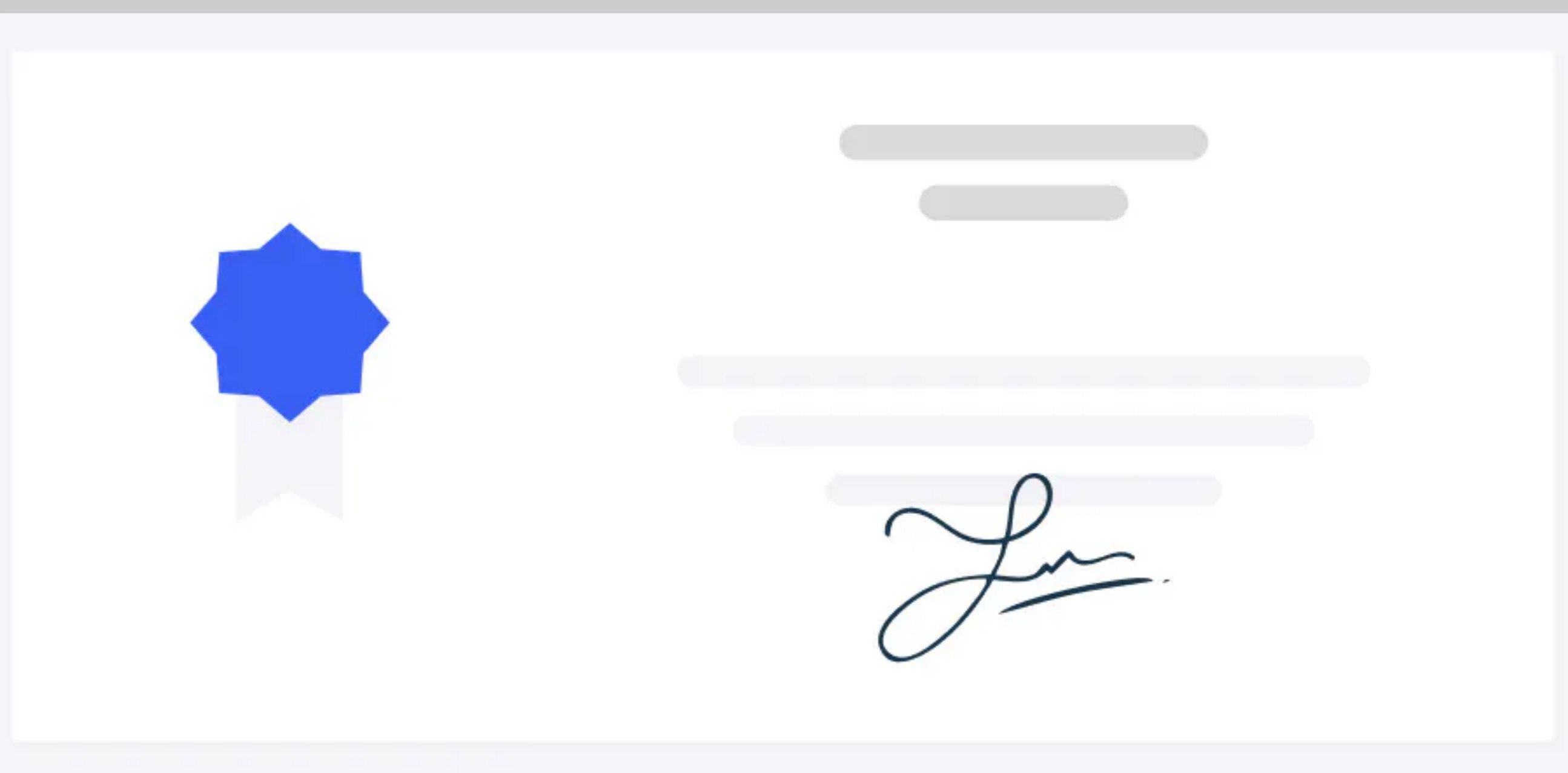
Course Overview

System design questions have increasingly become an integral part of software engineering interviews. For senior engineers, the discussion around system design is considered even more important than solving a coding question. In a system design interview, you can show your real design skills and show how they will work with designing complex systems. It is a given that a good performance in system design interviews will get you a senior position and result in higher salaries.

This course presents the architectural review of famous distributed systems. The main goal is to extract out important design details that are relevant to system design interviews. The course also presents a list of system design patterns that constitute the common design problems and their solutions that different distributed systems have developed over time.



Start Reading



Certification on completing the course.

Download Certificate

Course Content

1. Introduction
1 chapter

- What Is This Course About?

2. Dynamo: How to design a key value store?
12 chapters

- Dynamo: Introduction
- High-Level Architecture
- Data Partitioning
- Replication
- Vector Clocks and Conflicting Data
- The Life of Dynamo’s put() & get() Operations
- Anti-entropy Through Merkle Trees
- Gossip Protocol
- Dynamo Characteristics and Criticism
- Summary: Dynamo
- Quiz: Dynamo
- Mock Interview: Dynamo

3. Cassandra: How to Design a Wide-column NoSQL Database?
12 chapters

- Cassandra: Introduction
- High-level Architecture
- Replication
- Cassandra Consistency Levels
- Gossiper
- Anatomy of Cassandra's Write Operation
- Anatomy of Cassandra's Read Operation
- Compaction
- Tombstones
- Summary: Cassandra
- Quiz: Cassandra
- Mock Interview: Cassandra

4. Kafka: How to Design a Distributed Messaging System?
13 chapters

- Messaging Systems: Introduction
- Kafka: Introduction
- High-level Architecture
- Kafka: Deep Dive
- Consumer Groups
- Kafka Workflow
- Role of ZooKeeper
- Controller Broker
- Kafka Delivery Semantics
- Kafka Characteristics
- Summary: Kafka
- Quiz: Kafka
- Mock Interview: Kafka

5. Chubby: How to Design a Distributed Locking Service?
14 chapters

- Chubby: Introduction
- High-level Architecture
- Design Rationale
- How Chubby Works
- File, Directories, and Handles
- Locks, Sequencers, and Lock-delays
- Sessions and Events
- Master Election and Chubby Events
- Caching
- Database
- Scaling Chubby
- Summary: Chubby
- Quiz: Chubby
- Mock Interview: Chubby

6. GFS: How to Design a Distributed File System Storage?
15 chapters

- Google File System: Introduction
- High-level Architecture
- Single Master and Large Chunk Size
- Metadata
- Master Operations
- Anatomy of a Read Operation
- Anatomy of a Write Operation
- Anatomy of an Append Operation
- GFS Consistency Model and Snapshotting
- Fault Tolerance, High Availability, and Data Integrity
- Garbage Collection
- Criticism on GFS
- Summary: GFS
- Quiz: GFS
- Mock Interview: GFS

7. HDFS: How to Design File Storage System?
12 chapters

- Hadoop Distributed File System: Introduction
- High-level Architecture
- Deep Dive
- Anatomy of a Read Operation
- Anatomy of a Write Operation
- Data Integrity & Caching
- Fault Tolerance
- HDFS High Availability (HA)
- HDFS Characteristics
- Summary: HDFS
- Quiz: HDFS
- Mock Interview: HDFS

8. BigTable: How to Design a Wide Column Storage System?
15 chapters

- BigTable: Introduction
- BigTable Data Model
- System APIs
- Partitioning and High-level Architecture
- SSTable
- GFS and Chubby
- Bigtable Components
- Working with Tablets
- The Life of BigTable's Read & Write Operations
- Fault Tolerance and Compaction
- BigTable Refinements
- BigTable Characteristics
- Summary: BigTable
- Quiz: BigTable
- Mock Interview: BigTable

9. System Design Patterns
21 chapters

- Introduction: System Design Patterns
- 1. Bloom Filters
- 2. Consistent Hashing
- 3. Quorum
- 4. Leader and Follower
- 5. Write-ahead Log
- 6. Segmented Log
- 7. High-Water Mark
- 8. Lease
- 9. Heartbeat
- 10. Gossip Protocol
- 11. Phi Accrual Failure Detection
- 12. Split Brain
- 13. Fencing
- 14. Checksum
- 15. Vector Clocks
- 16. CAP Theorem
- 17. PACELC Theorem
- 18. Hinted Handoff
- 19. Read Repair
- 20. Merkle Trees

10. Final Assessment
2 chapters

- Quiz I
- Quiz II

11. Appendix
1 chapter

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